

Downloading Python

It depends on which operating system you're using. Downloading Python is an easy process. This is one of Python's main goal. To be easy to run, easy to install. You can download python from different resources. The official Python website is the go-to to download python. It has the [download links](#) for all operating systems. It has detailed release notes for differences between versions. You need to take a look at the release notes if you're upgrading your python version. It'll help you know the changes exactly.

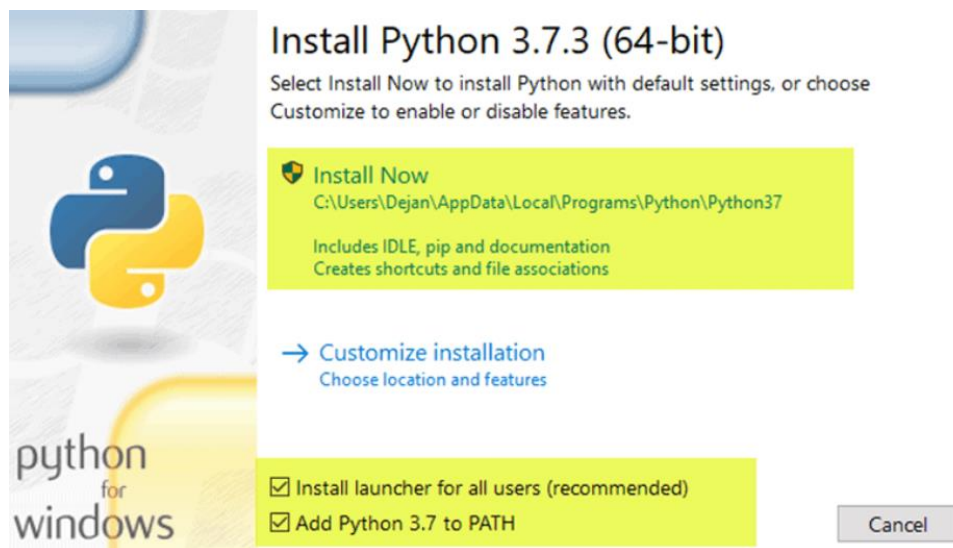
To download python, you need to

1. Go to the [download page on the official website](#).
2. Select your operating system. ([Windows](#), [Mac](#), [Unix](#))
3. Click on the exact version you want to download. (Latest version is recommended)

Your download will start automatically. Now it's time to install Python.

If you want to install python Globally on windows. You should follow these steps. After downloading Python. In your downloads, you'll have a .exe file.

You should follow the instruction for the installation in windows



When the installation is finished, you can confirm the installation by opening up Command Prompt and typing the following command:

```
C:\Users\User>python -V
Python 3.9.7
```

Installing pip

[pip](#) is the reference Python package manager. It's used to install and update packages. You'll need to make sure you have the latest version of pip installed.

The Python installers for Windows include pip. You can make sure that pip is up-to-date by running:

```
C:\Users\User>py -m pip install --upgrade pip
Requirement already satisfied: pip in c:\users\user\appdata\local\programs\python\python39\lib\site-packages (22.0.4)
Collecting pip
  Downloading pip-22.1.2-py3-none-any.whl (2.1 MB)
     2.1/2.1 MB 7.2 MB/s eta 0:00:00
Installing collected packages: pip
  Attempting uninstall: pip
    Found existing installation: pip 22.0.4
    Uninstalling pip-22.0.4:
      Successfully uninstalled pip-22.0.4
Successfully installed pip-22.1.2
```

Creating a virtual environment

[venv](#) (for Python 3) and [virtualenv](#) (for Python 2) allow you to manage separate package installations for different projects. They essentially allow you to create a “virtual” isolated Python installation and install packages into that virtual installation. When you switch projects, you can simply create a new virtual environment and not have to worry about breaking the packages installed in the other environments. It is always recommended to use a virtual environment while developing Python applications.

If you are using Python 3.3 or newer, the [venv](#) module is the preferred way to create and manage virtual environments. venv is included in the Python standard library and requires no additional installation.

To create a virtual environment, go to your project's directory and run venv:

```
C:\Users\User>py -m venv c:\users\user\workshop
```

Activating a virtual environment: Before you can start installing or using packages in your virtual environment, you'll need to *activate* it.

```
C:\Users\User>workshop\Scripts\activate.bat
(workshop) C:\Users\User>
```

If you want to switch projects or otherwise leave your virtual environment, simply run:

```
(workshop) C:\Users\User>deactivate  
C:\Users\User>
```

How to Install packages

Now that you're in your virtual environment you can install packages. Let's install the library **scikit-image**

To install the current **scikit-image** you'll need at least Python 3.6. If your Python is older, pip will find the most recent compatible version.

But first: Update pip

```
(workshop) C:\Users\User>python -m pip install -U pip  
Requirement already satisfied: pip in c:\users\user\workshop\lib\site-packages (21.2.3)  
Collecting pip  
  Using cached pip-22.1.2-py3-none-any.whl (2.1 MB)  
Installing collected packages: pip  
  Attempting uninstall: pip  
    Found existing installation: pip 21.2.3  
    Uninstalling pip-21.2.3:  
      Successfully uninstalled pip-21.2.3  
Successfully installed pip-22.1.2
```

And install scikit-image

```
(workshop) C:\Users\User>pip install -U scikit-image  
Collecting scikit-image
```

What is scikit-image?

scikit-image is a collection of algorithms for image processing. It is available free of charge and free of restriction.

[GitHub - scikit-image/scikit-image: Image processing in Python](https://github.com/scikit-image/scikit-image)

Install required packages

pip install pandas

pandas is a fast, powerful, flexible and easy to use open source data analysis and manipulation tool, built on top of the Python programming language.

pip install plotly

Plotly's Python graphing library makes interactive, publication-quality graphs.

pip install matplotlib

Matplotlib is a comprehensive library for creating static, animated, and interactive visualizations in Python. Matplotlib makes easy things easy and hard things possible.

pip install jupyter notebook

Free software, open standards, and web services for interactive computing across all programming languages

To run jupyter notebook

Open Command Prompt and typing the following command: jupyter notebook

Next you can add your virtual environment to Jupyter by typing:

(Remember to activate **virtual environment**)

```
C:\Users\User>workshop\Scripts\activate.bat
(workshop) C:\Users\User>python -m ipykernel install --user --name=workshop
```